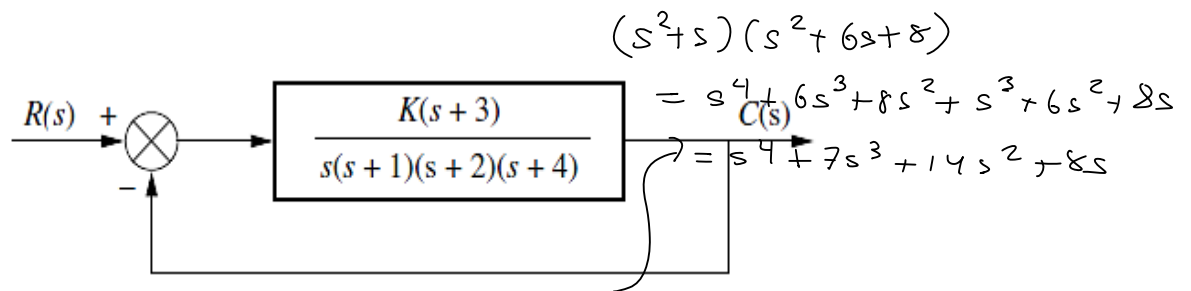


## Assignment 12 (ELEC 341 L12\_RLEexamples)

### Problem 1:

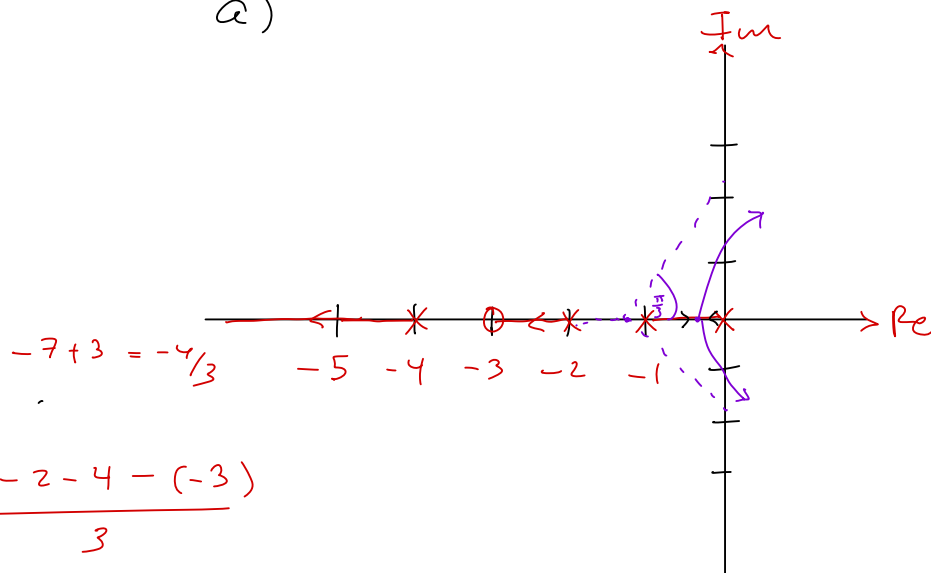
Sketch the root locus for the following system:



poles at  $s = 0, -1, -2, -4$   
zeroes at  $s = -3$ .

$$L(s) = \frac{N(s)}{D(s)} = \frac{s+3}{s^4 + 7s^3 + 14s^2 + 8s}$$

a)



$$r = 3$$

$$\sigma_a = \frac{0 - 1 - 2 - 4 - (-3)}{3} = -4/3$$

$$\sigma_a = -4/3 = -1.33$$

$$b) \theta_a = \frac{\pi}{r} (2k+1)$$

$$k=0 \quad k=1 \quad k=2$$

$$\theta_1 = \frac{\pi}{3} \quad \theta_2 = \pi \quad \theta_3 = \frac{5\pi}{3}$$

$$\frac{dL}{ds} \Rightarrow N'(s)D(s) - D'(s)N(s) = 0$$

$$(1)(s^4 + 7s^3 + 14s^2 + 8s) - (4s^3 + 21s^2 + 28s + 8)(s+3) = 0$$

$$s^4 + 7s^3 + 14s^2 + 8s - (4s^4 + 33s^3 + 91s^2 + 92s + 24) = 0$$

$$-3s^4 - 26s^3 - 77s^2 - 84s - 24 = 0$$

$$3s^4 + 26s^3 + 77s^2 + 84s + 24 = 0$$

used calc  $\left\{ \begin{array}{l} s_1 = -1.6097 \text{ not on RL} \\ s_2 = -0.4349 \end{array} \right.$

Problem 2:

Given the unity feedback system of Figure , where

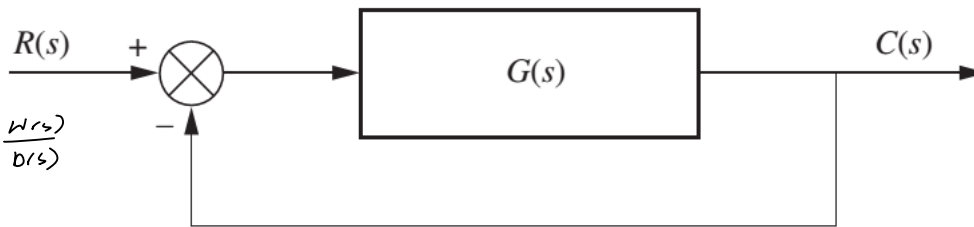
$$G(s) = \frac{K(s+2)}{s(s+1)(s+3)(s+5)}$$

do the following:

- Sketch the root locus.
- Find the asymptotes.
- Find the value of gain that will make the system marginally stable.
- Find the value of gain for which the closed-loop transfer function will have a pole on the real axis at  $-0.5$ .

$$\begin{aligned} & \sim (s^2+s)(s^2+8s+15) \\ & = s^4 + 8s^3 + 15s^2 + s^3 + 8s^2 + 15s \\ & = s^4 + 9s^3 + 23s^2 + 15s \end{aligned}$$

$$(1+G(s)=0) (s+0.5) \Rightarrow \text{solve for } K \text{ using RHC.}$$



FIGURE

$$b. \sigma_a = \frac{\sum \text{pole} - \sum \text{zero}}{r}, \quad r=3$$

$$\sigma_a = \frac{-1-3-5+2}{3}$$

$$\sigma_a = -7/3 = -2.667$$

$$\theta_a = \frac{\pi}{r}(2k+1)$$

$$k=0 \quad k=1 \quad k=2$$

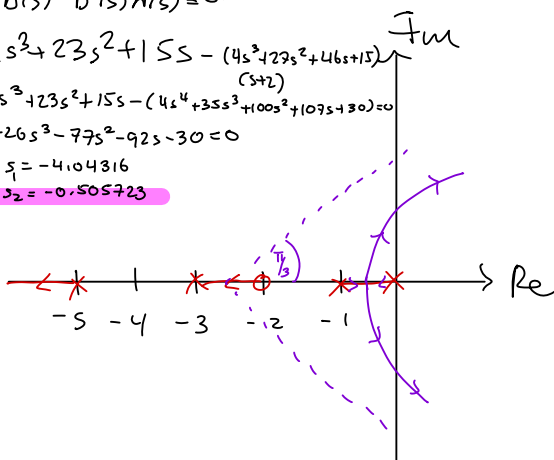
$$\theta_1 = \frac{\pi}{3} \quad \theta_2 = \pi \quad \theta_3 = 5\pi/3$$

$$\begin{array}{r|rrrr} s^4 & 1 & 23 & 2K & \\ s^3 & 9 & 15+K & 0 & \\ s^2 & \frac{192-K}{9} & 2K & 0 & \\ s^1 & K_1 & 0 & 0 & \\ s^0 & 2K & & & \end{array} \quad \begin{aligned} x_1 &= \frac{(92-K)(15+K) - 2K(81)}{192-K} \\ 2880 + 192K - 15K - K^2 - 162K &> 0 \\ -K^2 + 15K + 2880 &> 0 \\ K^2 - 15K - 2880 &< 0 \\ K_1 &< 61.687175 \\ K_2 &> 46.687175 \end{aligned}$$

$$K = 61.687175$$

Is for marginal stability.

$$\begin{aligned} a. \quad \frac{dL}{ds} & \Rightarrow N'(s)D(s) - D'(s)N(s) = 0 \\ & \Rightarrow s^4 + 9s^3 + 23s^2 + 15s - (4s^3 + 27s^2 + 46s + 15) = 0 \\ & \Rightarrow s^4 + 9s^3 + 23s^2 + 15s - (4s^3 + 27s^2 + 46s + 15) = 0 \\ & \Rightarrow -3s^4 - 26s^3 - 77s^2 - 92s - 30 = 0 \\ & s_1 = -4.104316 \\ & s_2 = -0.505723 \end{aligned}$$



$$c. 1+G(s) = 0$$

$$1 + \frac{K(s+2)}{s(s+1)(s+3)(s+5)} = 0$$

$$s^4 + 9s^3 + 23s^2 + 15s + Ks + 2K = 0$$

$$s^4 + 9s^3 + 23s^2 + (15+K)s + 2K = 0$$

$$d. \quad K = \frac{-1}{L(s)} \Big|_{s=-0.5}$$

$$L(s) = \frac{s+2}{s(s+1)(s+3)(s+5)}$$

$$K = 1.875$$

$$L(-0.5) = \frac{-0.5+2}{(-0.5)(0.5)(2.5)(4.5)}$$

$$L(-0.5) = -0.5333$$